Introduction to Real Analysis – Homework 9

Fatou, dominated convergence, and signed series

1. **Where did the probability go?** For each n, let $p_n(k) \geq 0$ for $k \in \mathbb{N} = \{1, 2, \ldots\}$, with $\sum_{k=1}^{\infty} p_n(k) = 1$. Suppose $p_n(k) \to p(k)$ for each k.

   (a) Use Fatou's lemma to show that $c := \sum_{k=1}^{\infty} p(k) \leq 1$.

   (b) Given any $c \in [0, 1)$, construct an example in which the limiting mass is exactly c.

   (c) Prove that
   $$\lim_{n\to\infty} \sum_{k=1}^{\infty} |p_n(k) - p(k)| = 1 - c.$$

   Hint. Apply dominated convergence to $\min\{p_n(k), p(k)\}$.

2. **Differentiating an integral.** For $t \in \mathbb{R}$, set
   $$F(t) = \int_0^{\infty} e^{-x} \frac{\sin(tx)}{x}\, dx.$$

   (a) Prove that $F(t)$ is well-defined for every t. Use dominated convergence on the difference quotients to show that
   $$F'(t) = \int_0^{\infty} e^{-x} \cos(tx)\, dx.$$

   Hint. Use $|\sin a - \sin b| \leq |a - b|$.

   (b) Evaluate $F'(t)$ and deduce a formula for $F(t)$.

3. **Cancellation in a series of functions.** On $(0, 1)$ with Lebesgue measure, let
   $$h_n = 2^n \mathbf{1}_{(0,\, 2^{-n})} \quad (n \geq 0), \qquad u_k = h_{k-1} - h_k \quad (k \geq 1).$$

   (a) Find the pointwise sum $\sum_{k=1}^{\infty} u_k$. Compare its integral with $\sum_{k=1}^{\infty} \int_0^1 u_k(x)\, dx$.

   (b) Compute $\int_0^1 |u_k(x)|\, dx$. In the theorem on integrating signed series term by term, can the assumption $\sum_k \int |u_k| < \infty$ be replaced by $\sum_k |\int u_k| < \infty$?